

Raphael Ramos

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EDUCATION

University of Alberta

Bachelor of Science in Computer Engineering

Edmonton, AB

Sept. 2021 – June 2026

CERTIFICATIONS

Engineer-in-Training (EIT), APEGA — 2026

TECHNICAL SKILLS

Languages: Python, C, C++, Java, HTML, VHDL, MATLAB, ARM, MIPS

Embedded & Hardware: STM32, STM32CubeIDE, Raspberry Pi Pico 2W, Xilinx Zybo Z7 FPGA, Edge Impulse, FreeRTOS, GDB, JTAG, Vivado, LTSpice, Cadence Virtuoso, WaveForms, KiCad, PCB Design

Protocols: I2C, SPI, UART, PWM

Cloud & Backend: AWS IoT Core, Lambda, DynamoDB, Firebase, MQTT/TLS

Tools & Libraries: Git, GitHub, Docker, Android Studio, VS Code, Linux, NumPy, Pandas, Scikit-learn, OpenMP

EXPERIENCE

Product Development Intern

May 2024 – Aug. 2024

TURBINE-X Energy Inc.

Nisku, AB

- Performed system-level testing and validation for fully integrated prototypes to ensure specification compliance.
- Calibrated advanced electrical, optical, and sensor equipment ensuring precise measurements across test campaigns.
- Built and brought up servers for internal data sharing, including physical construction and digital configuration.
- Created technical documentation for testing equipment covering training materials, specifications, and test conditions.

Software Deployment Intern

July 2023 – Sept. 2023

Vertical City

Edmonton, AB

- Updated Linux-based software from Ubuntu 18 to Ubuntu 20 across 250+ elevator and lobby screens.
- Troubleshoot LTE signal processing with antennas, optimizing network connectivity for uninterrupted communication.
- Executed hardware re-wiring on elevator TVs and collaborated with supervisors and building technicians to coordinate deployment.

PROJECTS

Continuing Care Home Activity Monitor | C++, MicroPython, Edge Impulse, AWS IoT, React Native Jan. – Apr. 2026

- Built a non-invasive elder care monitoring system across 3 subsystems: stove detection, water usage, and appliance tracking.
- Deployed edge ML on a Raspberry Pi Pico 2W via Edge Impulse — 97.73% accuracy, sub-10-second response.
- Implemented full AWS IoT pipeline — MQTT/TLS, DynamoDB, Lambda anomaly detection, and Firebase push notifications.
- Delivered a caregiver React Native mobile app with real-time alerts and activity history; prototype under \$200 CAD.

RP2040 Motor-Controller Carrier Board | KiCad, PCB Design, Schematic Capture

2026

- Designing a custom 2-layer PCB to replace a breadboard prototype (Pico, H-bridge, pot, LED ring, encoder) — schematic capture and footprints in KiCad; layout, ground pour, and JLCPCB fab to follow.

Anomaly Detection System | Python, Scikit-learn, Pandas, Matplotlib

Apr. 2025

- Built a detection pipeline using statistical, distance-based (k-NN, DBSCAN), and ML (One-Class SVM) techniques.
- Applied preprocessing for missing value handling, normalization, and stratified train-test-validation splits.
- Evaluated models via precision, recall, F1-score, ROC/PR curves; documented end-to-end in a reproducible Jupyter notebook.

Event Lottery System | Java, Firebase, Android Studio

Sept. – Nov. 2024

- Built an Android app with a fair lottery-based sign-up system, Firebase Firestore backend, and local push notifications.
- Applied OOP design principles (CRC cards, UML) with comprehensive JUnit testing for scalability and maintainability.

16-bit CPU Design | VHDL, Vivado, Xilinx Zybo Z7 FPGA

Nov. 2024

- Designed a 16-bit CPU with controller-datapath architecture: ALU, register file, accumulators, and tri-state buffer.
- Developed an FSM executing 13 custom instructions; implemented a bitwise multiplier as a validation test program.
- Resolved double-access load discrepancies through rigorous debugging and simulation in Vivado.

Digital IC Design | Cadence Virtuoso, CMOS, SPICE, WaveForms

Fall 2024

- Designed and simulated a complete CMOS digital system in Cadence Virtuoso: parameterized inverter, NAND gates, 4-to-1 MUX, TSPC flip-flop, and 2-bit shift register.
- Integrated a thermometer-to-binary encoder with shift registers and MUX as a complete end-to-end digital system.
- Performed transient/DC analysis, transistor sizing, and process-corner/temperature and dynamic-power characterization.

VOLUNTEER

Vice President, Academics

Apr. 2025 – Present

Computer Engineering Club, University of Alberta

Edmonton, AB

- Lead academic programming for 100+ members including workshops, study sessions, and exam preparation events.
- Coordinate with faculty and student groups to connect members with academic resources and opportunities.

Ground Station Software Team

Sept. 2023 – Sept. 2024

AlbertaSat, University of Alberta

Edmonton, AB

- Contributed to ground station software development for CubeSat operations and reliable data transmission.
- Collaborated with technical teams on implementation tasks and participated in regular development work sessions.